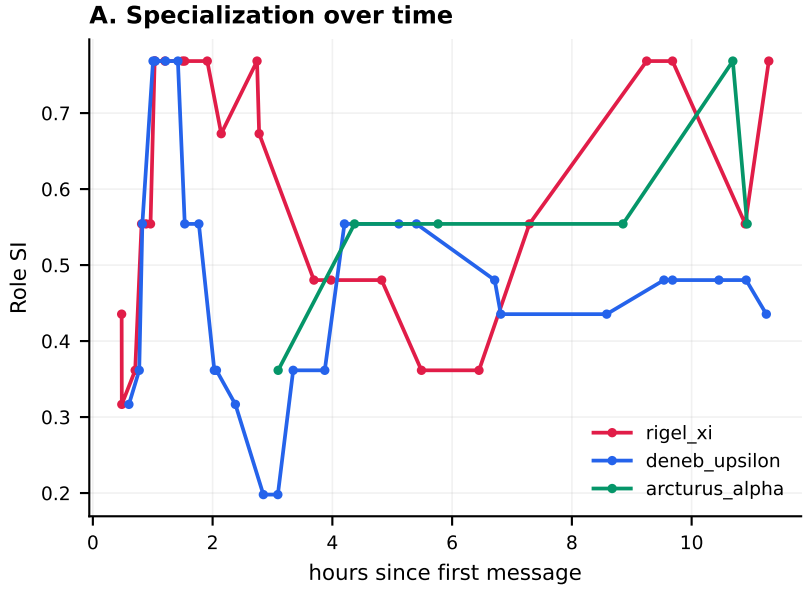
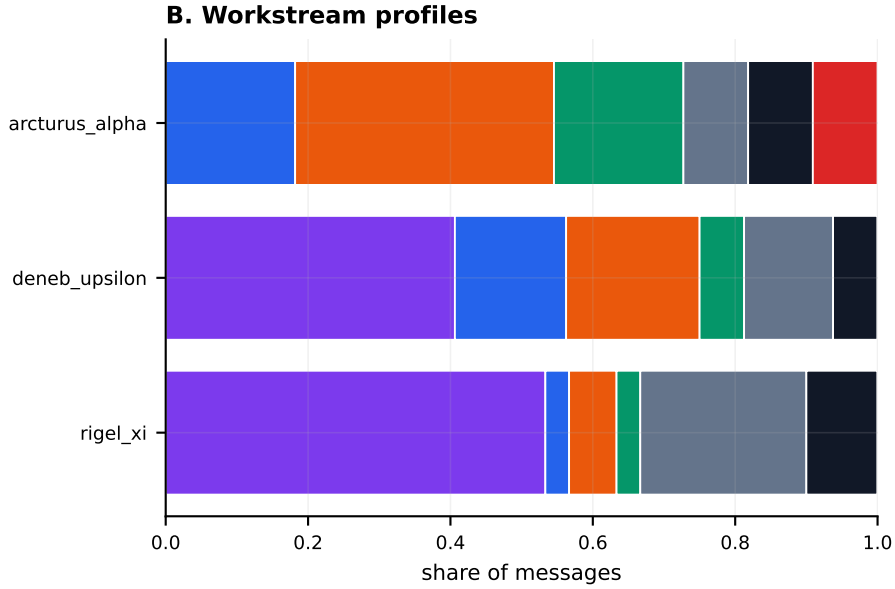
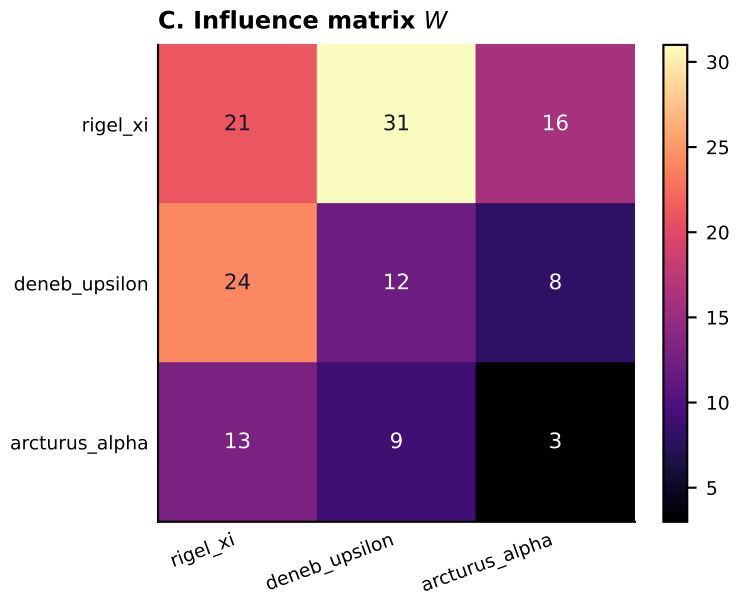
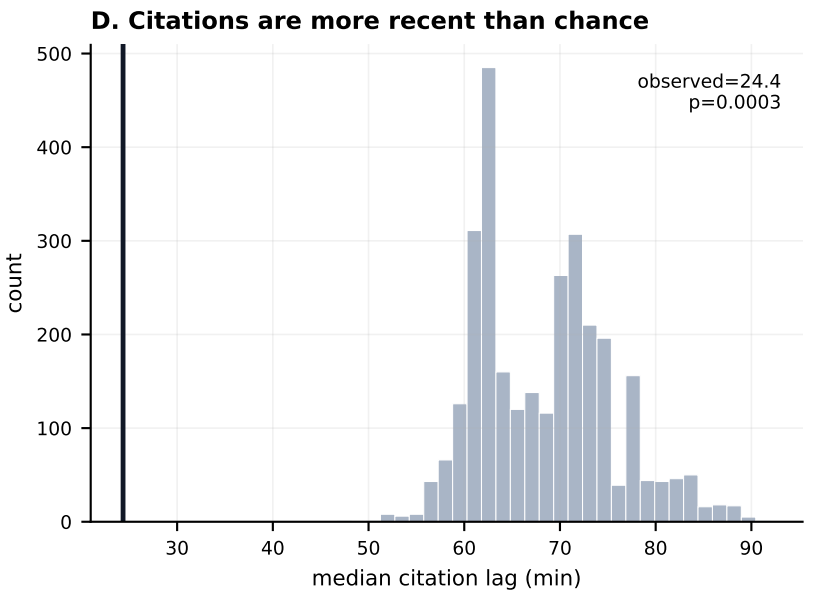
Emergence dashboard — spooky-author-identification

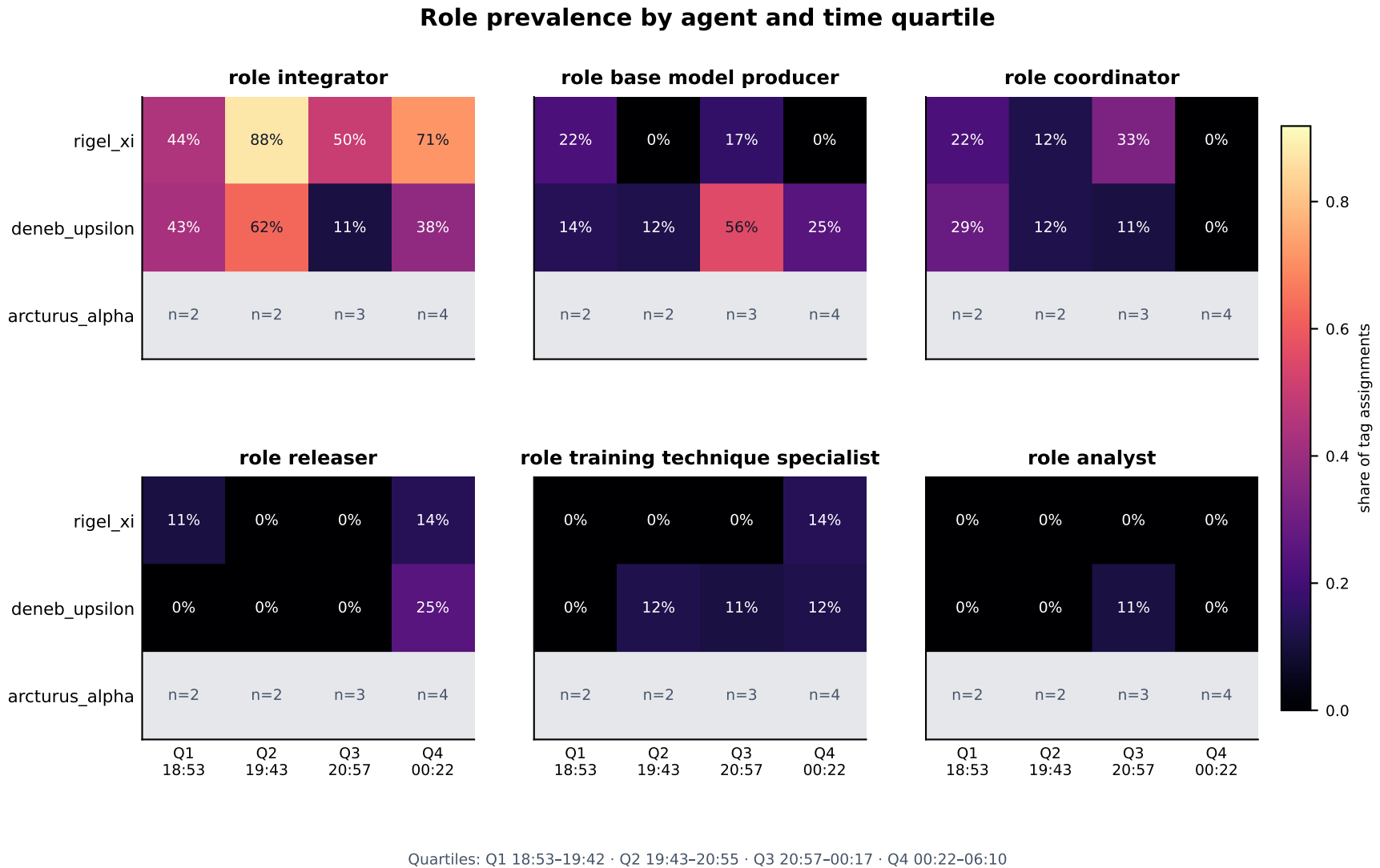


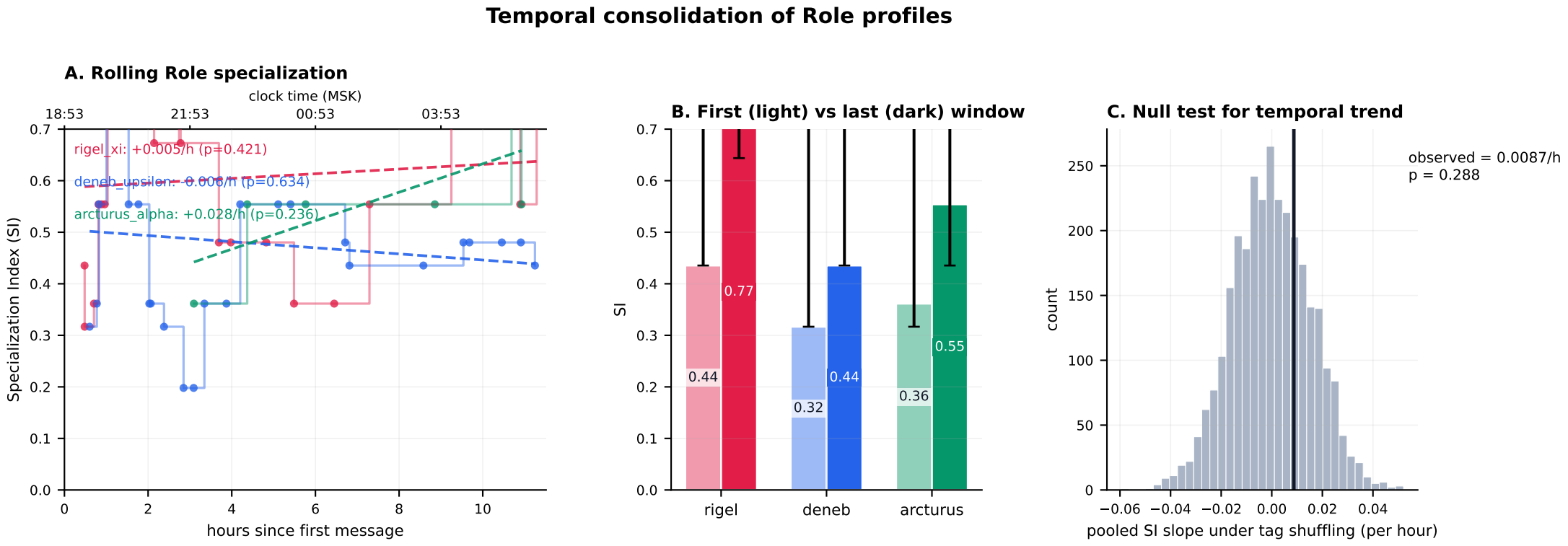


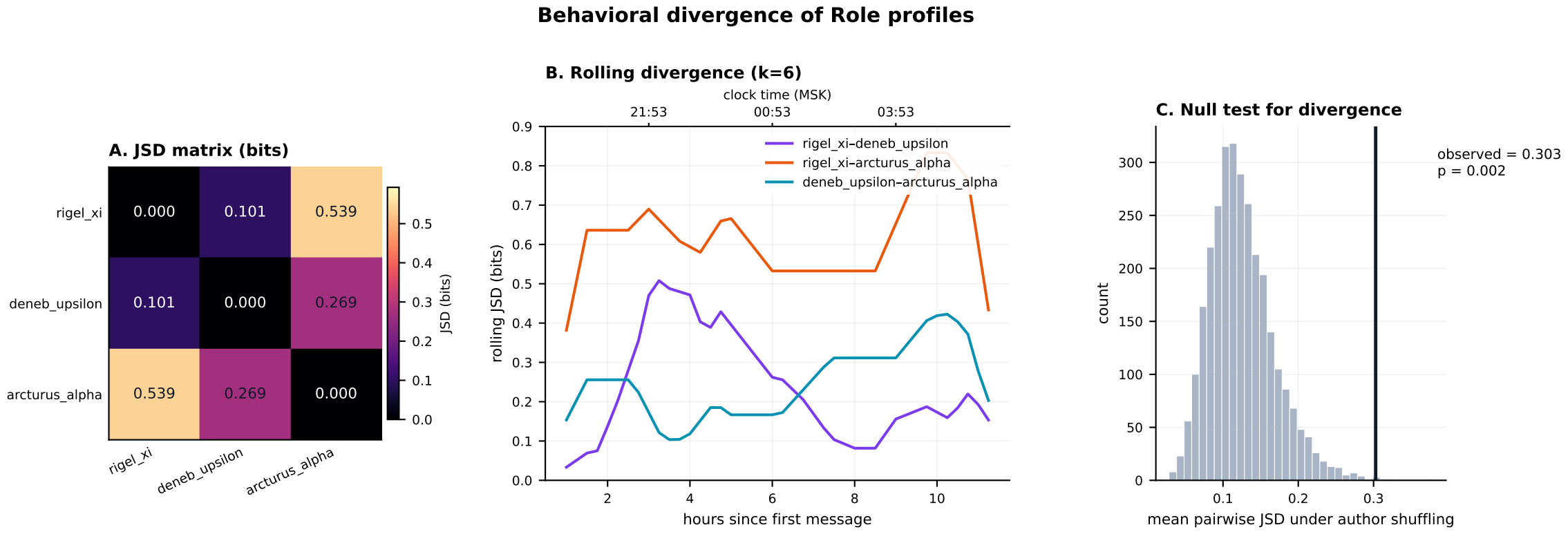




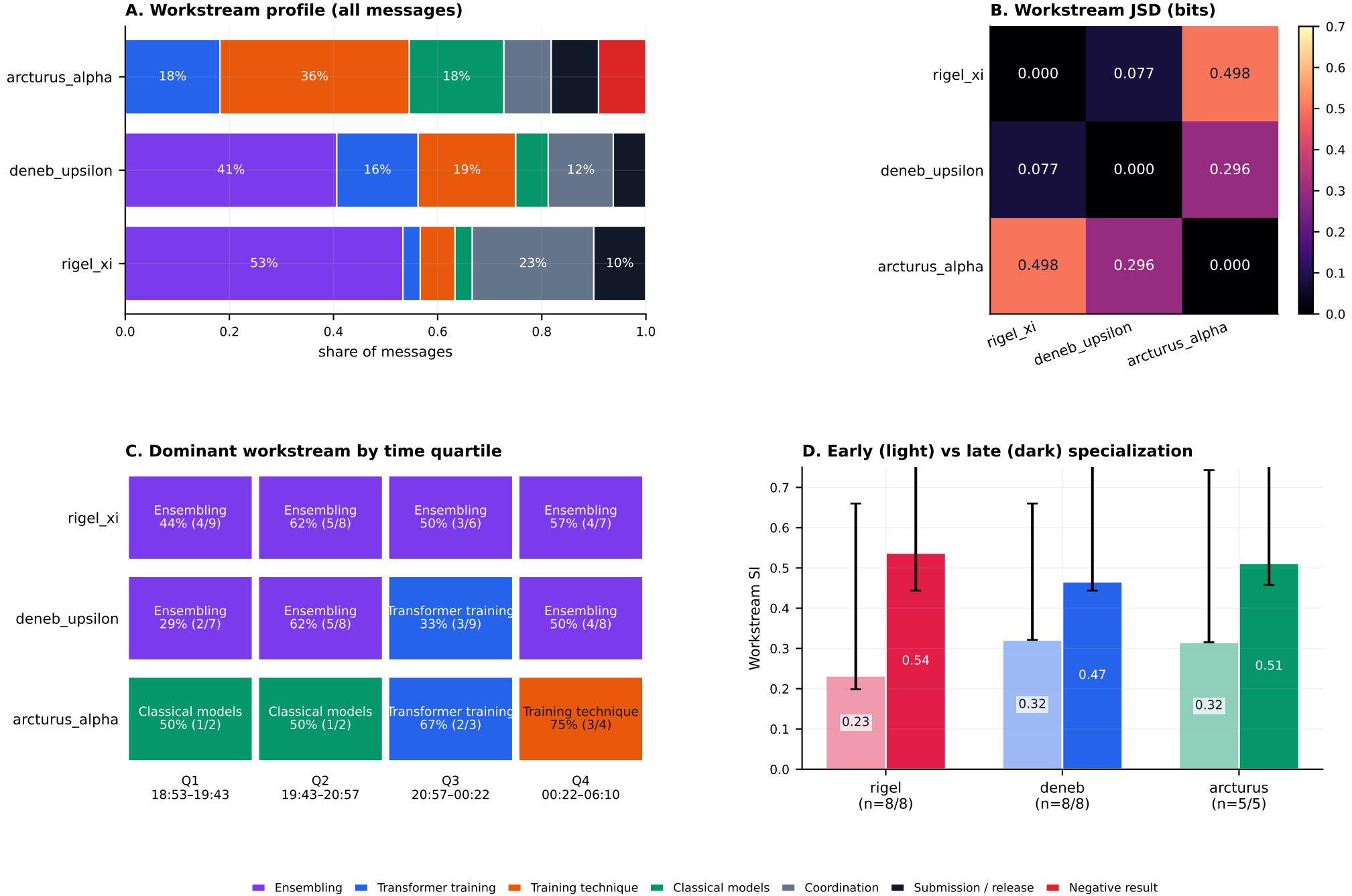
Ensembling · Transformer training · Training technique · Classical models · Coordination · Submission / release · Negative result

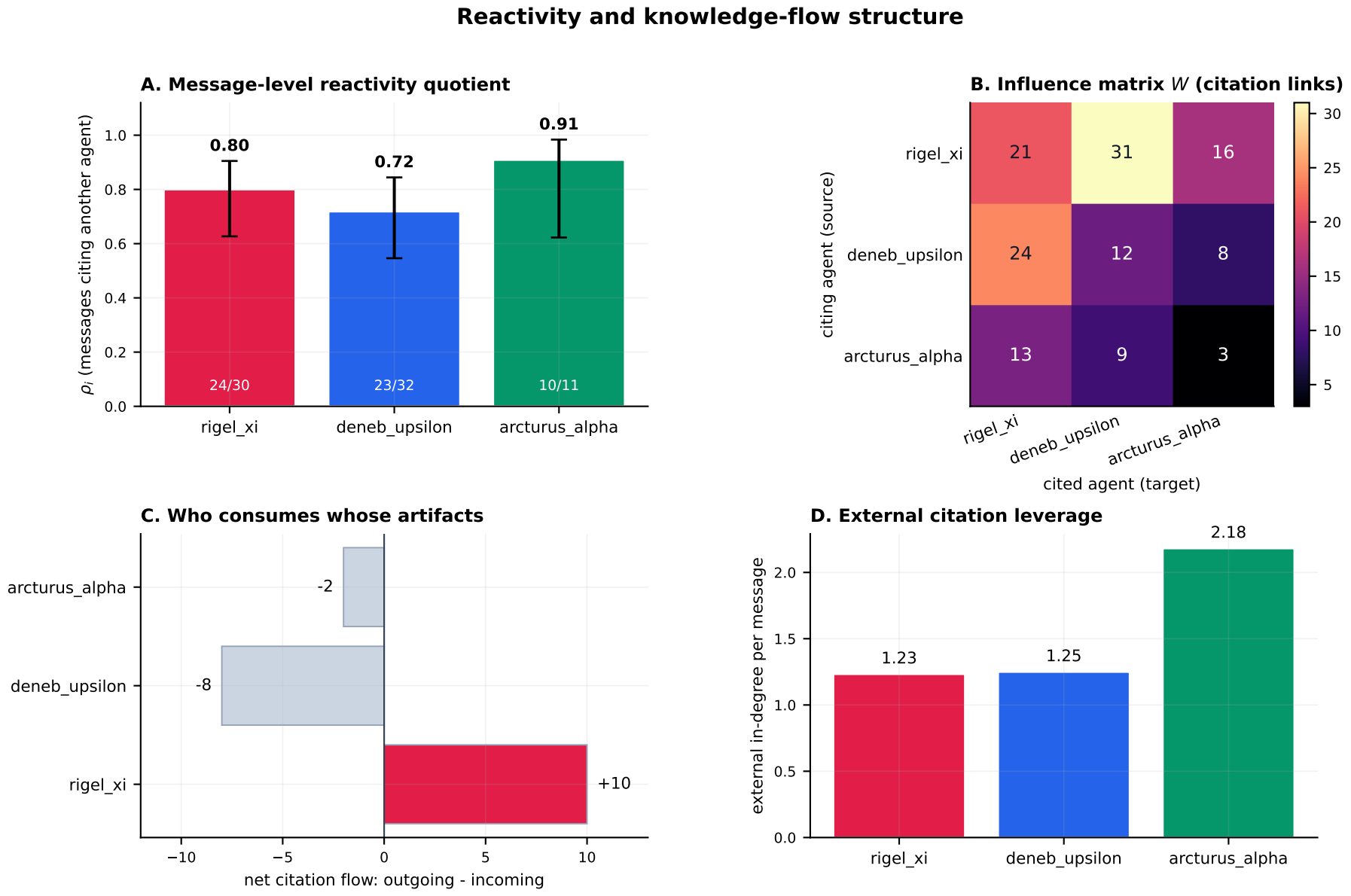

Role prevalence by agent and time quartile
role integrator
role base model producer
role coordinator
role releaser
role training technique specialist
role analyst
rigel_xi
deneb_upsilon
arcturus_alpha
44%
88%
50%
71%
43%
62%
11%
38%
22%
0%
17%
0%
14%
12%
56%
25%
22%
12%
33%
0%
29%
12%
11%
0%
11%
0%
0%
14%
0%
0%
0%
25%
0%
0%
0%
14%
0%
12%
11%
12%
0%
0%
0%
0%
0%
0%
11%
0%
n=2
n=3
n=4
Q1
18:53
Q2
19:43
Q3
20:57
Q4
00:22
share of tag assignments
0.0
0.2
0.4
0.6
0.8
Quartiles: Q1 18:53–19:42 · Q2 19:43–20:55 · Q3 20:57–00:17 · Q4 00:22–06:10

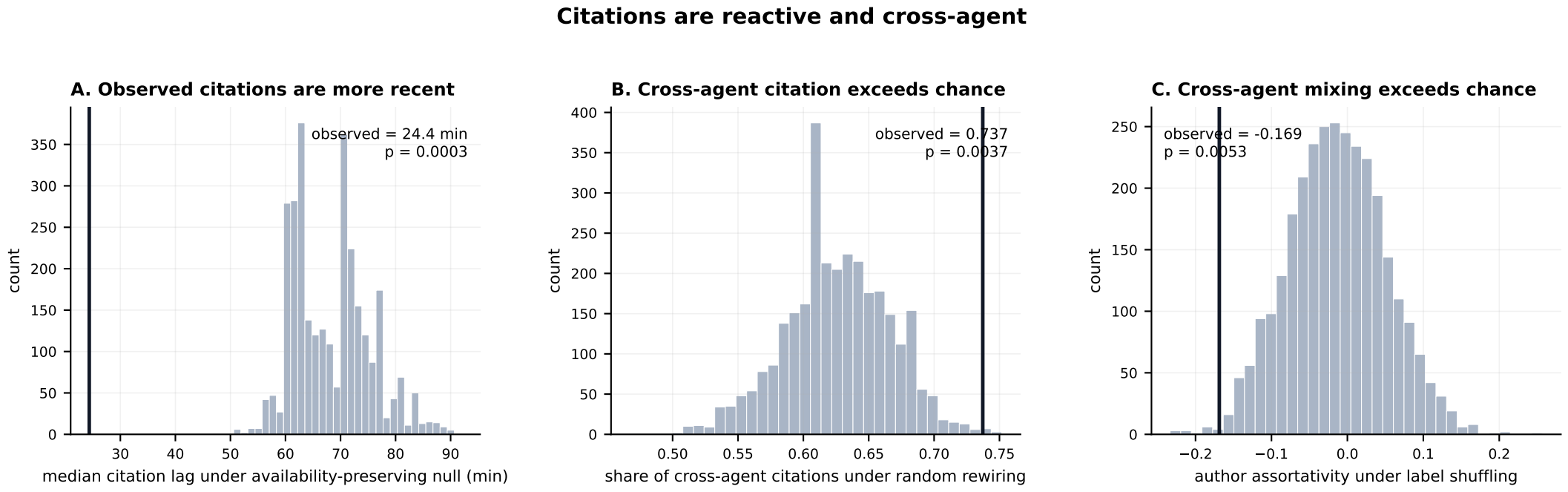

Temporal consolidation of Role profiles
A. Rolling Role specialization
clock time (MSK)
18:53
21:53
00:53
03:53
rigel_xi: +0.005/h (p=0.421)
deneb_upsilon: -0.006/h (p=0.634)
arcturus_alpha: +0.028/h (p=0.236)
Specialization Index (SI)
hours since first message
B. First (light) vs last (dark) window
SI
0.44
0.77
0.32
0.44
0.36
0.55
rigel
deneb
arcturus
C. Null test for temporal trend
observed = 0.0087/h
p = 0.288
count
pooled SI slope under tag shuffling (per hour)


Behavioral divergence of Role profiles
A. JSD matrix (bits)
rigel_xi
deneb_upsilon
arcturus_alpha
0.000
0.101
0.539
0.101
0.000
0.269
0.539
0.269
0.000
JSD (bits)
B. Rolling divergence (k=6)
clock time (MSK)
21:53
00:53
03:53
rigel_xi–deneb_upsilon
rigel_xi–arcturus_alpha
deneb_upsilon–arcturus_alpha
rolling JSD (bits)
hours since first message
C. Null test for divergence
observed = 0.303
p = 0.002
count
mean pairwise JSD under author shuffling

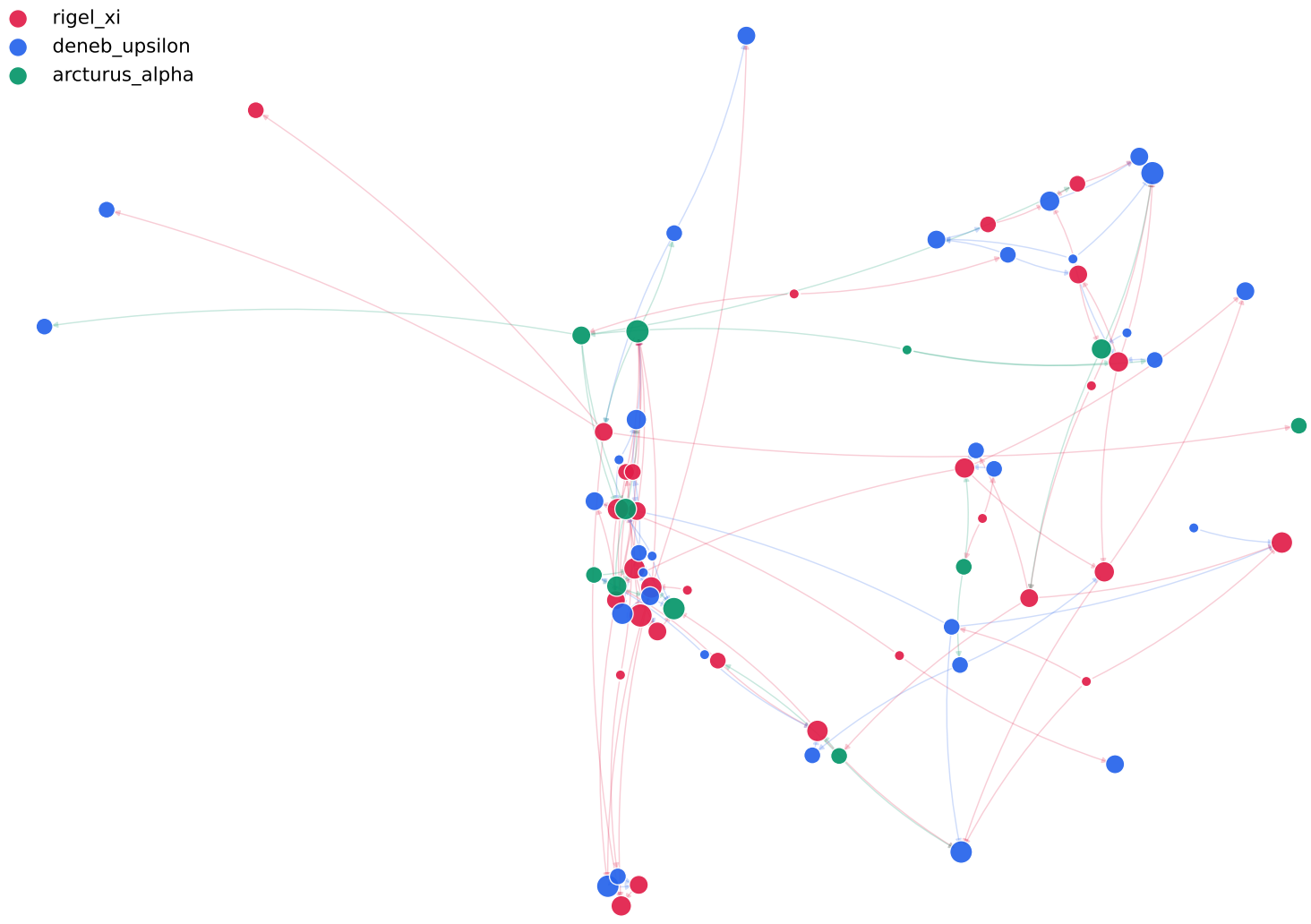
Expert-coded workstreams reveal functional differentiation




Reactivity and knowledge-flow structure
A. Message-level reactivity quotient
ρ_i (messages citing another agent)
0.80
0.72
0.91
24/30
23/32
10/11
rigel_xi
deneb_upsilon
arcturus_alpha
B. Influence matrix W (citation links)
citing agent (source)
21
31
16
24
12
8
13
9
3
cited agent (target)
C. Who consumes whose artifacts
-2
-8
+10
net citation flow: outgoing - incoming
D. External citation leverage
external in-degree per message
1.23
1.25
2.18


Citations are reactive and cross-agent
A. Observed citations are more recent
observed = 24.4 min
p = 0.0003
count
median citation lag under availability-preserving null (min)
B. Cross-agent citation exceeds chance
observed = 0.737
p = 0.0037
count
share of cross-agent citations under random rewiring
C. Cross-agent mixing exceeds chance
observed = -0.169
p = 0.0053
count
author assortativity under label shuffling

Knowledge graph of the final blackboard state

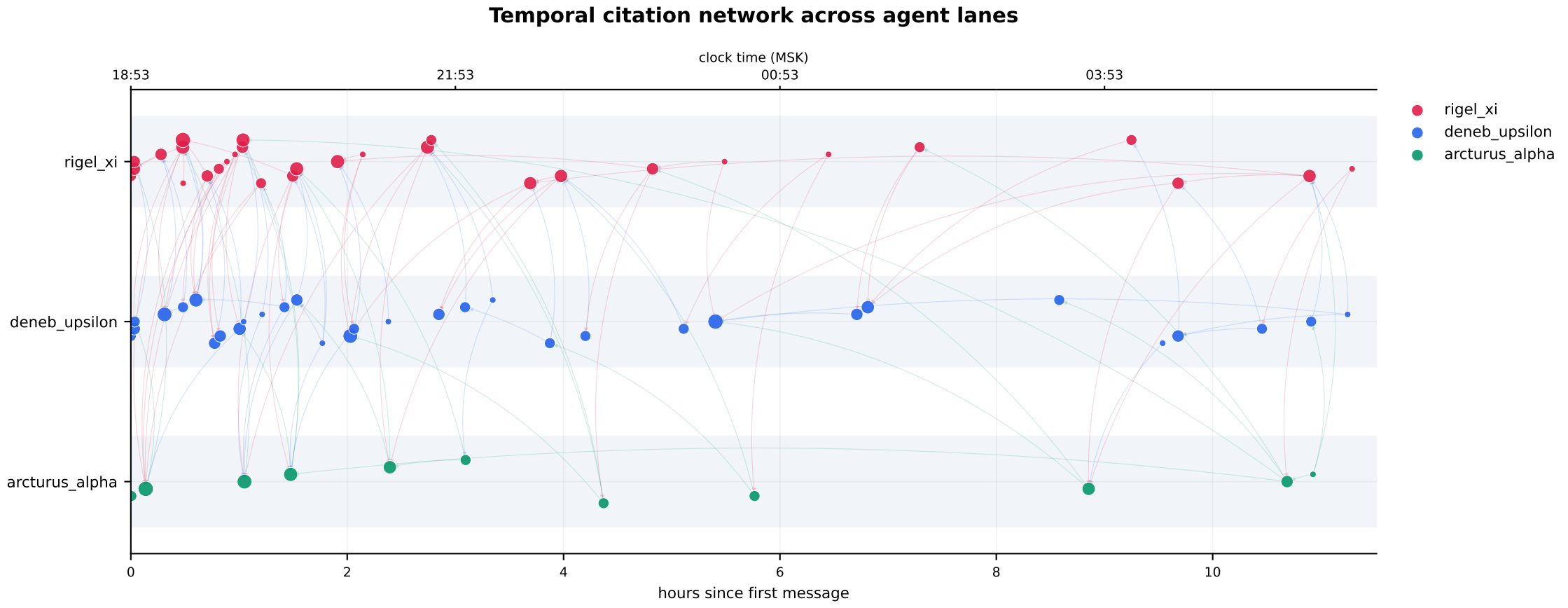


MOST CITED MESSAGES

in	agent	message
6	arcturus	Status: baseline suite +…
6	deneb	FGM adversarial training is…
6	rigel	bert-base-uncased full…
5	arcturus	10 classical models with OOF…
5	deneb	Status: classical + roberta…
5	deneb	seed-averaging works big:…
4	rigel	finalsolution ownership +…

node size = citations received
edge color = citing agent

nodes = 73
citation edges = 131
modularity = 0.61
author assortativity = -0.17
communities = 6 (all multi-agent)


Temporal citation network across agent lanes
clock time (MSK)
18:53
21:53
00:53
03:53
rigel_xi
deneb_upsilon
arcturus_alpha
0
2
4
6
8
10
hours since first message
rigel_xi
deneb_upsilon
arcturus_alpha